General Chemistry I
Recitation 13
Final Review

Siyuan Chen

Dec. 29

Final Announcement

课程序号	课程名称	上课人数	考试日期	考试时间	考试地点	主考教师
CHEM1103.01	普通化学 I	65	2026-01-09	10:30-12:30	教学中心 202	米启兮
CHEM1103.02	普通化学 I	45	2026-01-09	10:30-12:30	教学中心 203	宁志军
CHEM1103.03	普通化学 I	65	2026-01-09	10:30-12:30	教学中心 204	谢琎
CHEM1103.04	普通化学 I	65	2026-01-09	10:30-12:30	教学中心 302	杨帆
CHEM1103.05	普通化学 I	65	2026-01-09	10:30-12:30	教学中心 303	刘朋昕

- **When: 2026/01/09 (Friday), 120 minutes (10:30-12:30), week 17**
- **Where: Teaching Center 204**

ddl前提交期末课题！
作业/quiz有没批改/没有计分请联系助教！

考试注意事项：

1. **随身携带学生证/身份证**
2. **准备计算器！！！**
3. **提供参考纸**

Atomic
Symbol
Name
Weight

- **C** Solid
- **Hg** Liquid
- **H** Gas
- **Rf** Unknown

Metals: Alkali metals · Alkaline earth metals · Lanthanoids · Actinoids · Transition metals · Post-transition metals

Metalloids

Nonmetals: Reactive nonmetals · Noble gases

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15 Pnictogens	16 Chalcogens	17 Halogens	18
1	1 **H** Hydrogen 1.008																	2 **He** Helium 4.0026
2	3 **Li** Lithium 6.94	4 **Be** Beryllium 9.0122											5 **B** Boron 10.81	6 **C** Carbon 12.011	7 **N** Nitrogen 14.007	8 **O** Oxygen 15.999	9 **F** Fluorine 18.998	10 **Ne** Neon 20.180
3	11 **Na** Sodium 22.990	12 **Mg** Magnesium 24.305											13 **Al** Aluminium 26.982	14 **Si** Silicon 28.085	15 **P** Phosphorus 30.974	16 **S** Sulfur 32.06	17 **Cl** Chlorine 35.45	18 **Ar** Argon 39.948
4	19 **K** Potassium 39.098	20 **Ca** Calcium 40.078	21 **Sc** Scandium 44.956	22 **Ti** Titanium 47.867	23 **V** Vanadium 50.942	24 **Cr** Chromium 51.996	25 **Mn** Manganese 54.938	26 **Fe** Iron 55.845	27 **Co** Cobalt 58.933	28 **Ni** Nickel 58.693	29 **Cu** Copper 63.546	30 **Zn** Zinc 65.38	31 **Ga** Gallium 69.723	32 **Ge** Germanium 72.630	33 **As** Arsenic 74.922	34 **Se** Selenium 78.971	35 **Br** Bromine 79.904	36 **Kr** Krypton 83.798
5	37 **Rb** Rubidium 85.468	38 **Sr** Strontium 87.62	39 **Y** Yttrium 88.906	40 **Zr** Zirconium 91.224	41 **Nb** Niobium 92.906	42 **Mo** Molybdenum 95.95	43 **Tc** Technetium (98)	44 **Ru** Ruthenium 101.07	45 **Rh** Rhodium 102.91	46 **Pd** Palladium 106.42	47 **Ag** Silver 107.87	48 **Cd** Cadmium 112.41	49 **In** Indium 114.82	50 **Sn** Tin 118.71	51 **Sb** Antimony 121.76	52 **Te** Tellurium 127.60	53 **I** Iodine 126.90	54 **Xe** Xenon 131.29
6	55 **Cs** Caesium 132.91	56 **Ba** Barium 137.33	57–71	72 **Hf** Hafnium 178.49	73 **Ta** Tantalum 180.95	74 **W** Tungsten 183.84	75 **Re** Rhenium 186.21	76 **Os** Osmium 190.23	77 **Ir** Iridium 192.22	78 **Pt** Platinum 195.08	79 **Au** Gold 196.97	80 **Hg** Mercury 200.59	81 **Tl** Thallium 204.38	82 **Pb** Lead 207.2	83 **Bi** Bismuth 208.98	84 **Po** Polonium (209)	85 **At** Astatine (210)	86 **Rn** Radon (222)
7	87 **Fr** Francium (223)	88 **Ra** Radium (226)	89–103	104 **Rf** Rutherfordium (267)	105 **Db** Dubnium (268)	106 **Sg** Seaborgium (269)	107 **Bh** Bohrium (270)	108 **Hs** Hassium (277)	109 **Mt** Meitnerium (278)	110 **Ds** Darmstadtium (281)	111 **Rg** Roentgenium (282)	112 **Cn** Copernicium (285)	113 **Nh** Nihonium (286)	114 **Fl** Flerovium (289)	115 **Mc** Moscovium (290)	116 **Lv** Livermorium (293)	117 **Ts** Tennessine (294)	118 **Og** Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 **La** Lanthanum 138.91	58 **Ce** Cerium 140.12	59 **Pr** Praseodymium 140.91	60 **Nd** Neodymium 144.24	61 **Pm** Promethium (145)	62 **Sm** Samarium 150.36	63 **Eu** Europium 151.96	64 **Gd** Gadolinium 157.25	65 **Tb** Terbium 158.93	66 **Dy** Dysprosium 162.50	67 **Ho** Holmium 164.93	68 **Er** Erbium 167.26	69 **Tm** Thulium 168.93	70 **Yb** Ytterbium 173.05	71 **Lu** Lutetium 174.97
7	89 **Ac** Actinium (227)	90 **Th** Thorium 232.04	91 **Pa** Protactinium 231.04	92 **U** Uranium 238.03	93 **Np** Neptunium (237)	94 **Pu** Plutonium (244)	95 **Am** Americium (243)	96 **Cm** Curium (247)	97 **Bk** Berkelium (247)	98 **Cf** Californium (251)	99 **Es** Einsteinium (252)	100 **Fm** Fermium (257)	101 **Md** Mendelevium (258)	102 **No** Nobelium (259)	103 **Lr** Lawrencium (266)

Homework7

1. Predict the geometry and number of unpaired spins in the following four-coordinate complexes: $[Ni(CN)_4]^{2-}$, $[NiCl_4]^{2-}$. Explain your reasoning.

2. The octahedral structure is not the only possible six-coordinate structure. Other possibilities include a planar hexagonal structure and a triangular prism structure. In the latter, the ligands are arranged in two parallel triangles, one lying above the metal atom and the other below the metal atom with its corners directly in line with the corners of the first triangle. Show that the existence of two and only two isomers of $[Co(NH_3)_4Cl_2]^+$ is evidence against both of these possible structures.

3. Is the coordination compound $[Co(NH_3)_6]Cl_2$ diamagnetic or paramagnetic?

4. In what directions do you expect the bond length and vibrational frequency of a free CO molecule to change when it becomes a CO ligand in a $Ni(CO)_4$ molecule? Explain your reasoning.

5. Molecular nitrogen (N_2) can act as a ligand in certain coordination complexes. Predict the structure of $[V(N_2)_6]$, which is isolated by condensing V with N_2 at 25 K. Is this compound diamagnetic or paramagnetic? What is the formula of the carbonyl compound of vanadium that has the same number of electrons?

Quiz

Quiz11

可用性：项目已不可用。上次可用日期为 2025-12-12 下午12:00。

1. Would you expect the following reaction to proceed in aqueous solution as written?

$$CuCl_2(aq) + S^{2-}(aq) \rightarrow CuS(s) + 2Cl^-(aq)$$

2. Suppose 0.010 mol of each of the following compounds is dissolved (separately) in 1.0 L water: $BaCl_2$, $K_4[Fe(CN)_6]$, $[Cr(NH_3)_4Cl_2]Cl$, $[Fe(NH_3)_3Cl_3]$. Rank the resulting four solutions in order of conductivity, from lowest to highest.

Quiz12

可用性：项目已不可用。上次可用日期为 2025-12-19 下午12:00。

1. The octahedral complex ion $[MnCl_6]^{3-}$ has more unpaired spins than the octahedral complex ion $[Mn(CN)_6]^{3-}$. How many unpaired electrons are present in each species? Explain. In each case, express the CFSE in terms of Δ_0

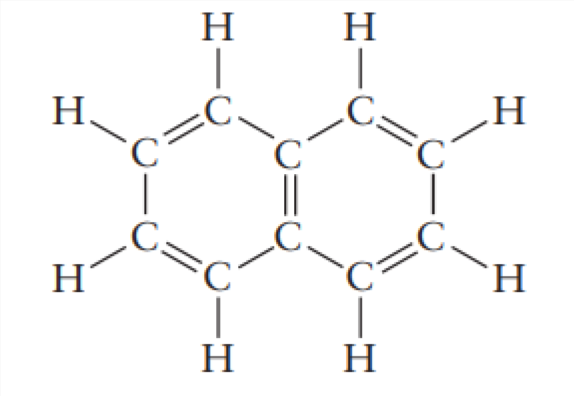
2. An aqueous solution of $Mn(NO_3)_2$ is very pale pink, but an aqueous solution of $K_4[Mn(CN)_6]$ is deep blue. Explain why the two differ so much in the intensities of their colors.

Quiz

Quiz13

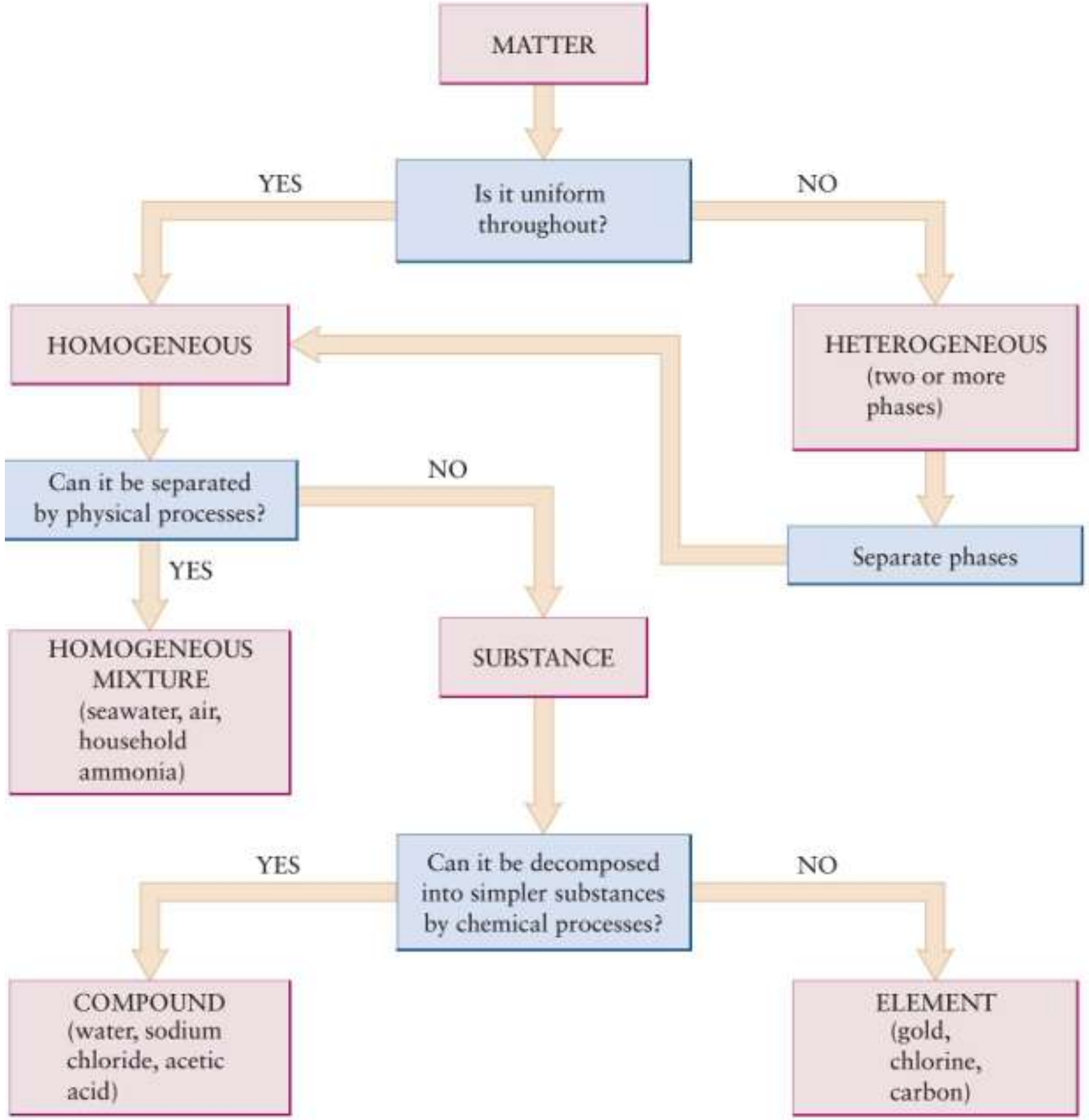
可用性: 项目已不可用。上次可用日期为 2025-12-26 下午12:00。

1. The naphthalene molecule has a structure that corresponds to two benzene molecules fused together: The p-electrons in this molecule are delocalized over the entire molecule. The wavelength of maximum absorption in the UV-visible part of the spectrum in benzene is 255 nm. Is the corresponding wavelength shorter or longer than 255 nm for naphthalene?



2. The bond dissociation energy of a typical C-Cl bond in a chlorofluorocarbon is approximately 330 kJ mol^{-1}. Calculate the maximum wavelength of light that can photodissociate a molecule of CCl_2F_2, breaking such a C-Cl bond.

The atom in modern chemistry



- **Laws of chemical combinations**

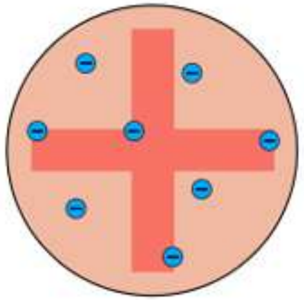
1. Law of Conservation of Mass

2. Law of Definite Proportions

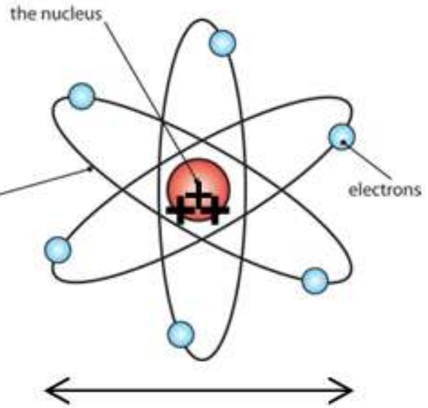
3. Law of Multiple Proportions

4. Law of Combining Volumes

5. Avogadro's Law



< 0.01 pm



~100 pm

Chemical formulas, equations, and reaction yields

- **The molecular formula is some whole-number multiple of the empirical formula.**

Nomenclature

- **From Avogadro's hypothesis (same number of molecules of two gases in same volume, temperature and pressure), the ratio of molar masses of two gaseous compounds is the same as the ratio of their densities.**

乙炔的密度

乙炔的摩尔质量

氧气的摩尔质量

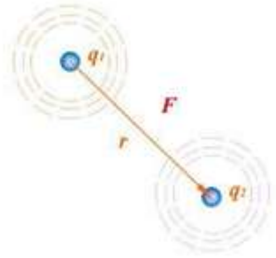
$$\text{Molar mass of welding gas} = \frac{1.06\ \text{g L}^{-1}}{1.31\ \text{g L}^{-1}}(32.0\ \text{g mol}^{-1}) = 25.9\ \text{g mol}^{-1}$$

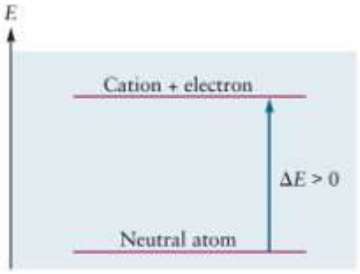
氧气的密度

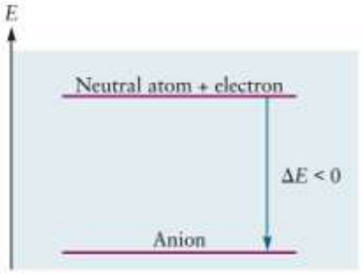
乙炔=C_2H_2

Chemical bonding: the classical description

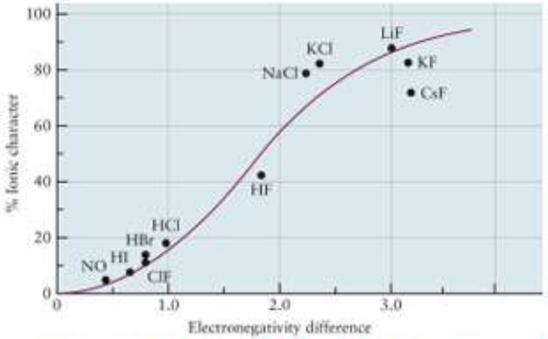
Coulomb's law (库仑定律)







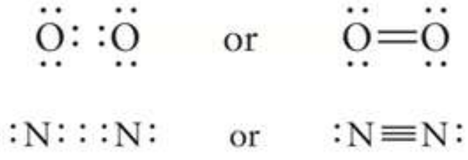
$X(g) \longrightarrow X^{+}(g) + e$ $\quad \Delta E = IE_1 \quad$ $X^{-}(g) \longrightarrow X(g) + e$



Ionic	Polar Covalent	Covalent
$\Delta\chi > 2.0$		$\Delta\chi < 0.4$

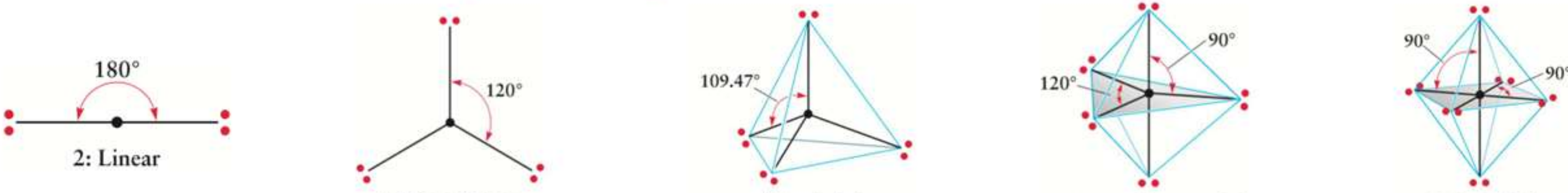
Lewis diagrams

- A shared pair electrons = a covalent bond
- A lone pair electrons = no bonding
- Each atom achieves its own noble-gas shell of electrons
- more than one pair may be shared (double bond, triple bond)



The Valence Shell Electron-Pair Repulsion Theory (VSEPR): 价层电子对互斥模型

$$SN = \begin{pmatrix}\text{number of atoms} \\ \text{bonded to central atom}\end{pmatrix} + \begin{pmatrix}\text{number of lone pairs} \\ \text{on central atom}\end{pmatrix}$$



2: Linear 3: Trigonal planar 4: Tetrahedral 5: Trigonal bipyramidal 6: Octahedral

Bonds and molecular structure

Formal charge (形式电荷): the charge an atom in a molecule would have if the electrons in its Lewis diagram were divided equally among the atoms that share them.

Lewis structure formula

Molecular Orbital	Lewis Dot Structure
Bonding orbital	Shared electron pair
Nonbonding orbital	Lone electron pair
Delocalized orbital	Resonance structures
Two electrons from one atom	Coordinate bond
	Formal charge

Formal charge = number of valence electrons

- number of electrons in lone pairs

- ½ (number of electrons in bonding pairs)

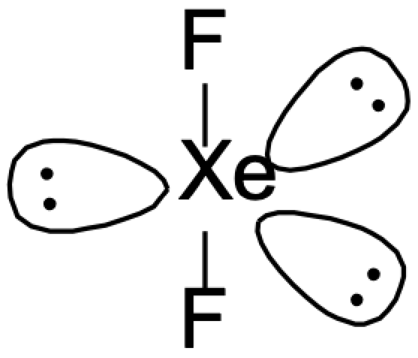
The less sum of FC(in absolute value), the more stable molecule is

Bonds and molecular structure

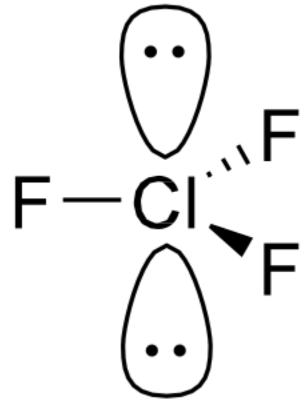
SN= Total amount of orbitals

SN of A	Molecular type X is a bonded atom; E is a lone pair	Predicted shape dsp²是什么形式？
2	AX_2	Linear (线形）
3	AX_3	Trigonal planar （三角形）
	AX_2E	Bent （弯曲形）
4	AX_4	Tetrahedral （四面体）
	AX_3E	Trigonal pyramidal （三角锥）
	AX_2E_2	Bent
5	AX_5	Trigonal bipyramid （三角双锥）
	AX_4E	Distorted see-saw （跷跷板）
	AX_3E_2	Distorted T （T字）
	AX_2E_3	Linear
6	AX_6	Octahedral （八面体）
	AX_5E	Square pyramidal （四角锥）
	AX_4E_2	Square planar （正方平面）

SN=3 $\rightarrow$ $sp^2(1+2=3)$



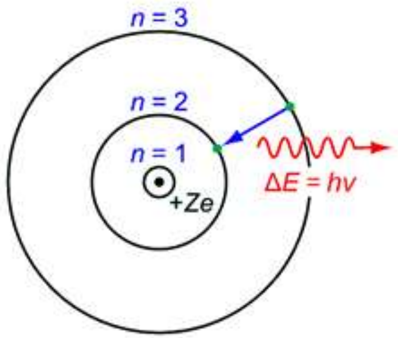
Why linear?



Why planar triangle?

Introduction to quantum mechanics

Bohr Model



$$L = m_e v r = n\frac{h}{2\pi} \qquad n = 1, 2, 3, \ldots$$

$$E_n = \frac{-Z^2 e^4 m_e}{8\epsilon_0^2 n^2 h^2} = -(2.18 \times 10^{-18}\,\mathrm{J})\frac{Z^2}{n^2} \qquad n = 1, 2, 3, \ldots$$

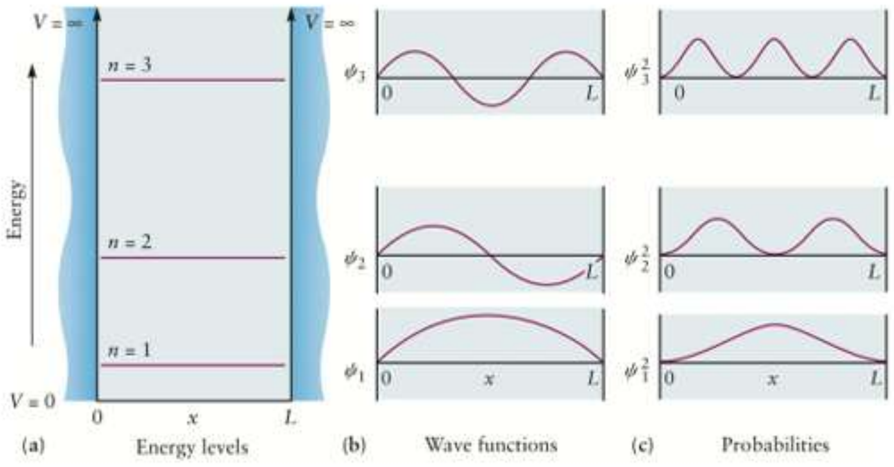
the Schrödinger equation $\hat{H}\Psi = i\hbar\frac{\partial}{\partial t}\Psi$

the Schrödinger equation (time-independent) $\hat{H}\psi = E\psi$

$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{p}^2}{2m} + V = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t). \qquad \sum_{i=1}^{n}\frac{\partial^2}{\partial x_i^2}$$

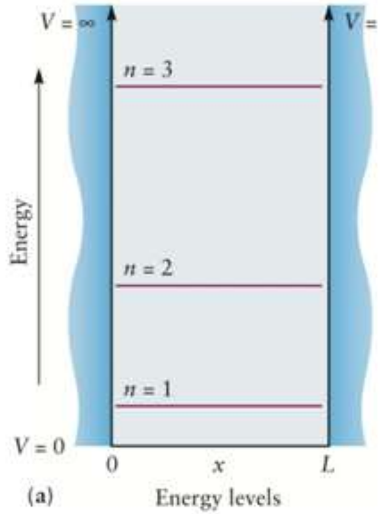
$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

the Schrödinger equation (time-independent, 1D)



The normalized wave function for the one-dimensional particle in a box is

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right) \quad n = 1, 2, 3, \ldots$$



$$E_n = \frac{n^2 h^2}{8mL^2} \qquad n = 1, 2, 3, \ldots$$

Probability of finding a particle is given by the square of the amplitude of the wave function

Probability: P(x)= |ψ(x)|^2

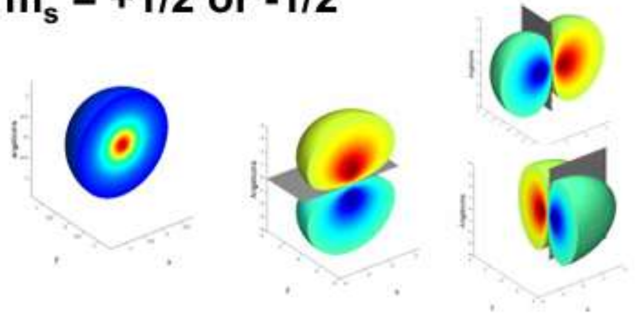
Quantum mechanics and atomic structure

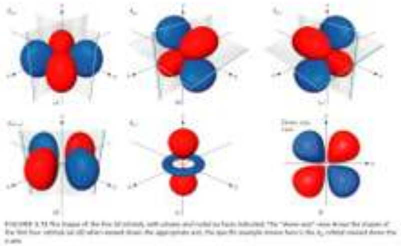
$n = 1, 2, 3, \ldots$ 主量子数

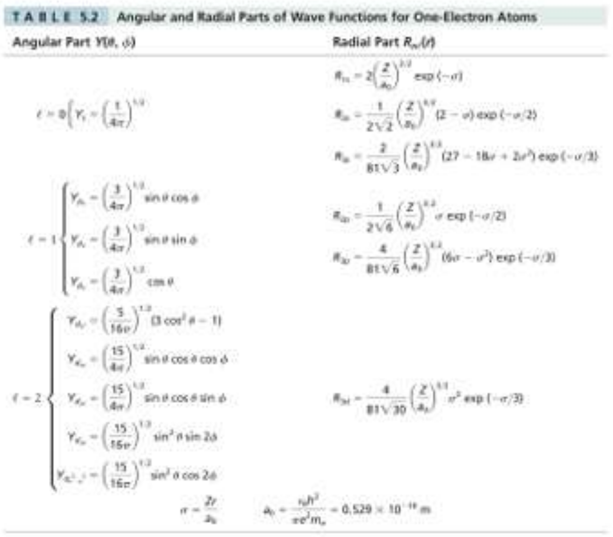
$\ell = 0, 1, \ldots, n-1$ 角量子数

$m = -\ell, -\ell+1, \ldots, 0, \ldots, \ell-1, \ell$ 磁量子数

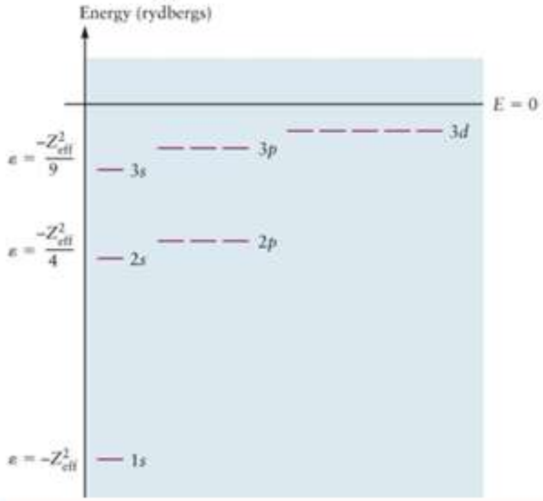
m_s = +1/2 or -1/2 自旋量子数

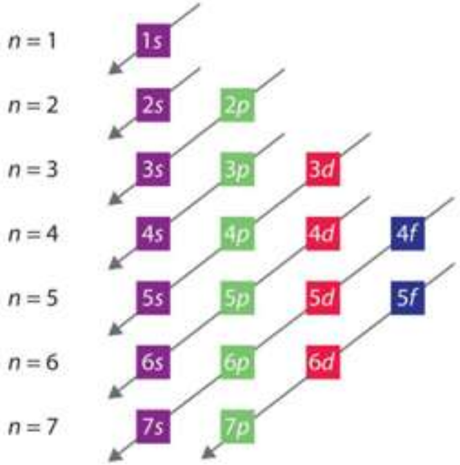






Hartree's method: Each electron moves in **an effective field** created by the nucleus and all the other electrons





The **Pauli exclusion principle**: **no two electrons in an atom can have the same set of four quantum numbers** (n, ℓ, m, m_s).

Hund's rules: when electrons are added to Hartree orbitals of equal energy, **a single electron enters each orbital** before a second one enters any orbital. In addition, the lowest energy configuration is the one **with parallel spins**.

Wave functions

$\ell = 0$ with s, $\ell = 1$ with p, $\ell = 2$ with d, $\ell = 3$ with f, $\ell = 4$ with g
1s
2s 2p
3s 3p 3d
4s 4p 4d 4f

TABLE 5.1 Allowed Values of Quantum Numbers for One-Electron Atoms

n	1	2		3		
ℓ	0	0	1	0	1	2
m	0	0	−1, 0, +1	0	−1, 0, +1	−2, −1, 0, +1, +2
Number of degenerate states for each ℓ	1	1	3	1	3	5
Number of degenerate states for each n	1	4		9		

Quantum mechanics and molecular structure

Linear Combination of Atomic Orbitals (LCAO，原子轨道线性组合)：

selecting sums and differences (linear combinations) of atomic orbitals to generate the best approximation to each type of molecular orbital.

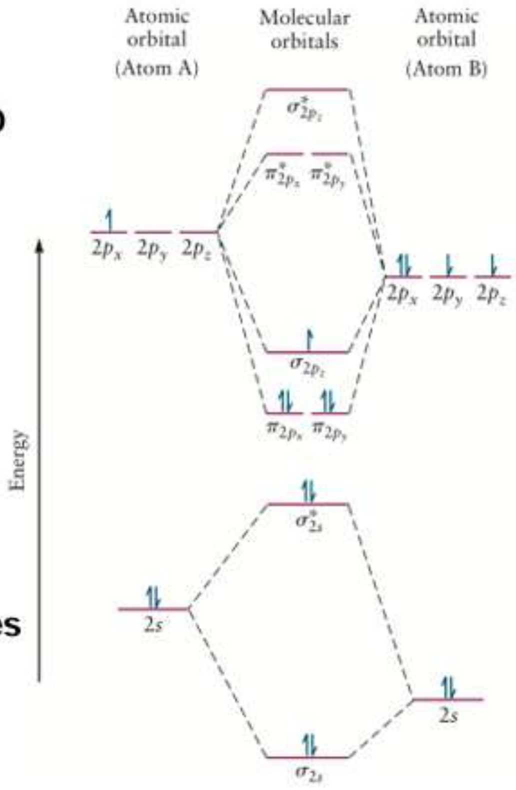
$$\psi_{\mathrm{MO}}^{\mathrm{el}} = \sigma_{g1s}(1)\sigma_{g1s}(2) = [1s^{\mathrm{A}}(1) + 1s^{\mathrm{B}}(1)][1s^{\mathrm{A}}(2) + 1s^{\mathrm{B}}(2)]$$

Correlation diagram (轨道相关图): the energy-level diagram within the LCAO approximation.

- Obey the Aufbau Principle

Rules of bonding

1. Maximum orbital overlap
2. Compatible orbital symmetries
3. Approximate atomic orbital energies



Valence bond (VB) model:

- Grew out of the qualitative Lewis electron-pair model in which the chemical bond is described as a pair of electrons that is localized between two atoms.
- The VB model constructs a wave function for the bond by assuming that each atom arrives with at least one unpaired electron in an AO.

$$\psi_{\mathrm{VB}}^{\mathrm{el}} = 1s^{\mathrm{A}}(1)1s^{\mathrm{B}}(2) + 1s^{\mathrm{A}}(2)1s^{\mathrm{B}}(1)$$

Promotion (激发) **Hybridization (杂化)**

TABLE 6.5

Molecular Shapes Predicted by the Valence Shell Electron-Pair Repulsion Theory and Rationalized by Orbital Hybridization

Molecule	Steric Number	Number of Lone Pairs	Orbital Hybridization	Predicted Geometry	Image	Example
AX_2	2	0	sp	Linear	 180°	BeH_2, CO_2
AX_3	3	0	sp^2	Trigonal planar	 120°	BF_3, SO_3
AX_2	3	1	sp^2	Bent		SO_2
AX_4	4	0	sp^3	Tetrahedral	 109.5°	CF_4, SO_4^{2-}
AX_3	4	1	sp^3	Trigonal pyramidal		NH_3, PF_3, $AsCl_3$
AX_2	4	2	sp^3	Bent		H_2O, H_2S, SF_2

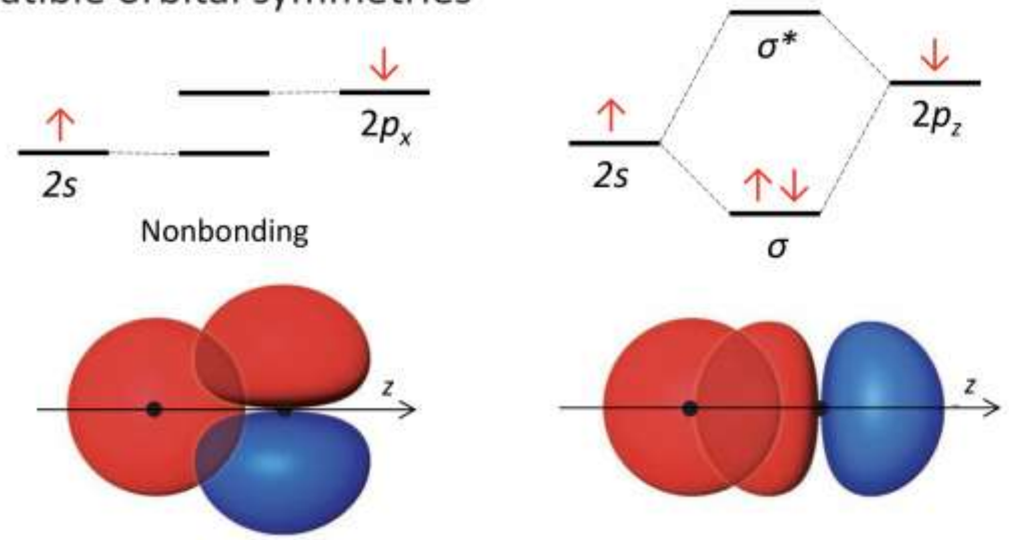
MO and molecular structure

MO theory

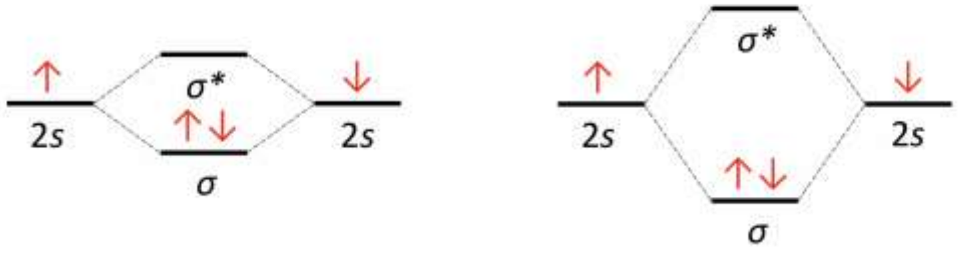
Molecular orbitals arise from allowed interactions between atomic orbitals (Atomic orbital overlap)
How to form efficient overlap:

- symmetries are compatible with each other.
- atomic orbitals are close in energy.
- the number of molecular orbitals formed must be equal to the number of atomic orbitals in the atoms being combined to form the molecule.

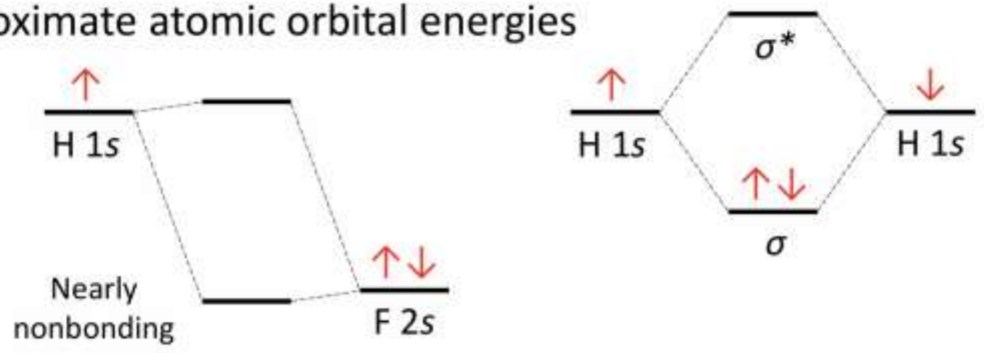
② Compatible orbital symmetries



① Maximum orbital overlap

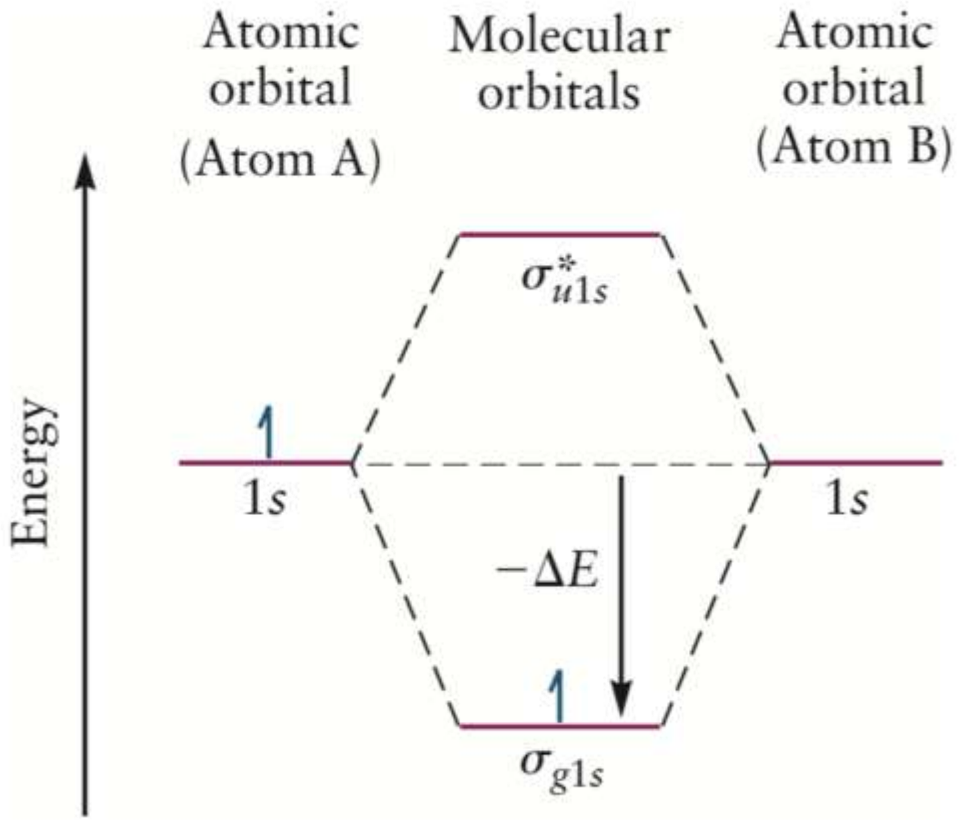


③ Approximate atomic orbital energies



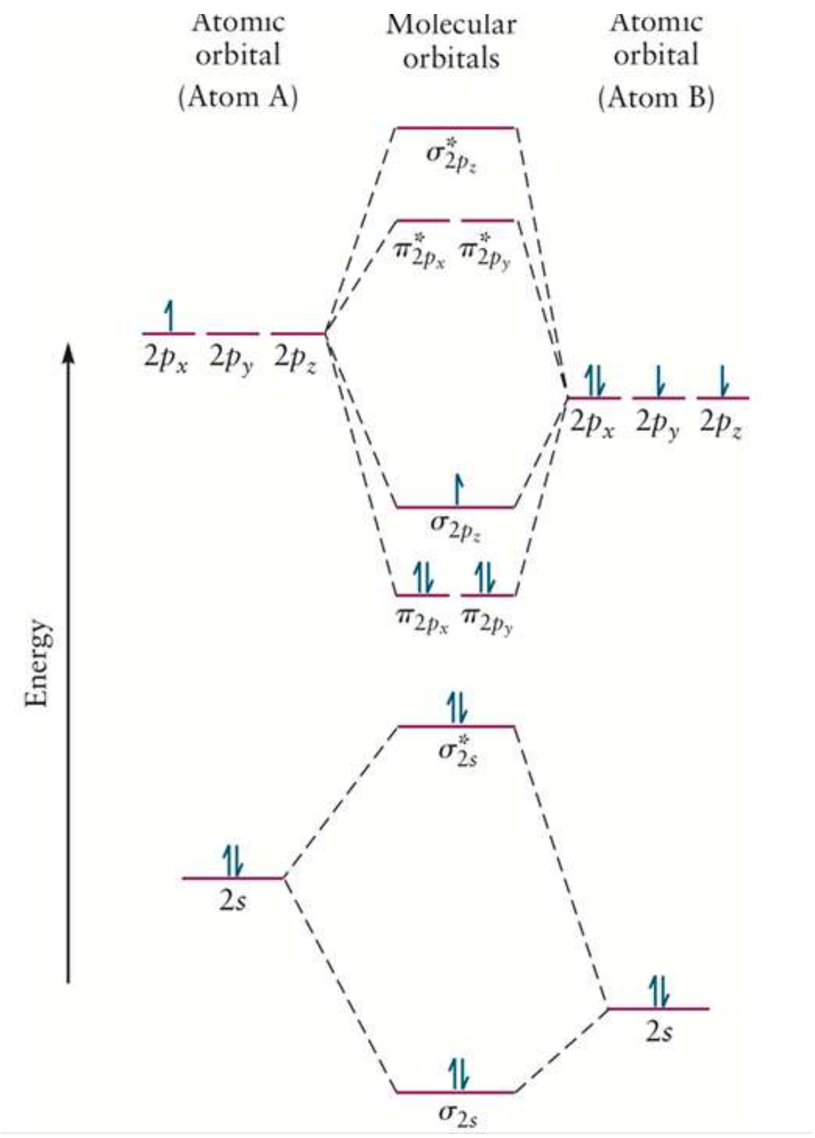
MO and molecular structure

How to fill the orbitals with electrons?



- Construct all molecular orbitals.
- Half bonding orbitals and half anti-bonding orbitals(what will happen if orbital number is odd?)
- Identify all valence electrons of the molecule.
- Populate the electrons in order of increasing energy (taking care to adhere to Hund's rule).

MO and molecular structure

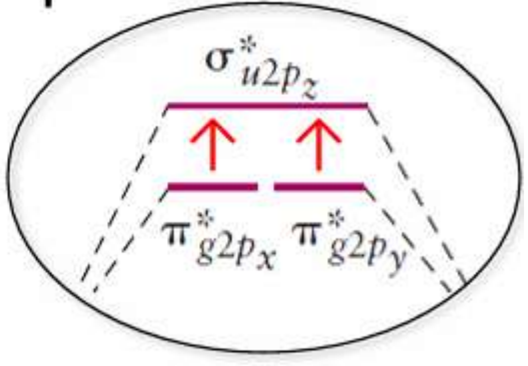


What can we learn from MO diagrams?

- Understand the bonding situation
- Deduce magnetic properties
- How molecules can change with ionization
- Bond order

$$\text{bond order} = \tfrac{1}{2}\,(\text{number of electrons in bonding molecular orbitals} - \text{number of electrons in antibonding molecular orbitals})$$

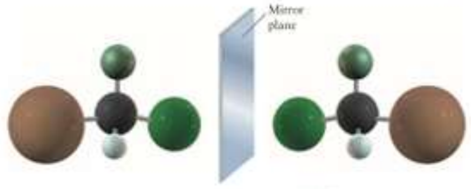
O_2 is paramagnetic due to its two unpaired electron



7. Bonding in organic molecules

- Geometrical isomers几何异构体/同分异构体: same molecular formula but a different bonding structure

- Optical isomerism 光学异构体: the two forms rotate the plane of polarized light in different directions



- 顺反异构体: cis (顺式) and trans (反式) forms

cis-2-butene *trans*-2-butene

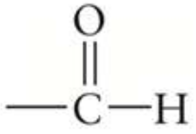
Alkanes (烷烃) vs. Alkenes (烯烃) vs. Alkynes (炔烃)

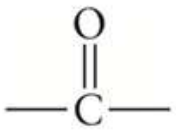
Alcohols (醇类): -OH functional group

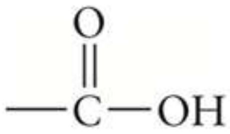
Phenols (苯酚类): -OH group is attached directly to an aromatic ring

Ethers (醚类): −O− functional group

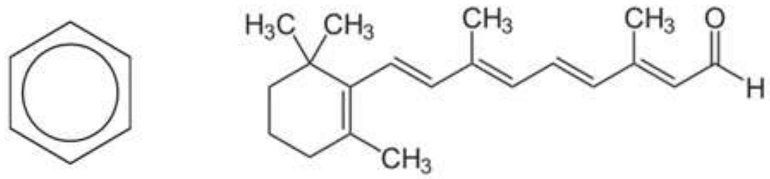
Aldehydes (醛类) vs. Ketones (酮类) vs. Carboxylic Acids (羧酸)





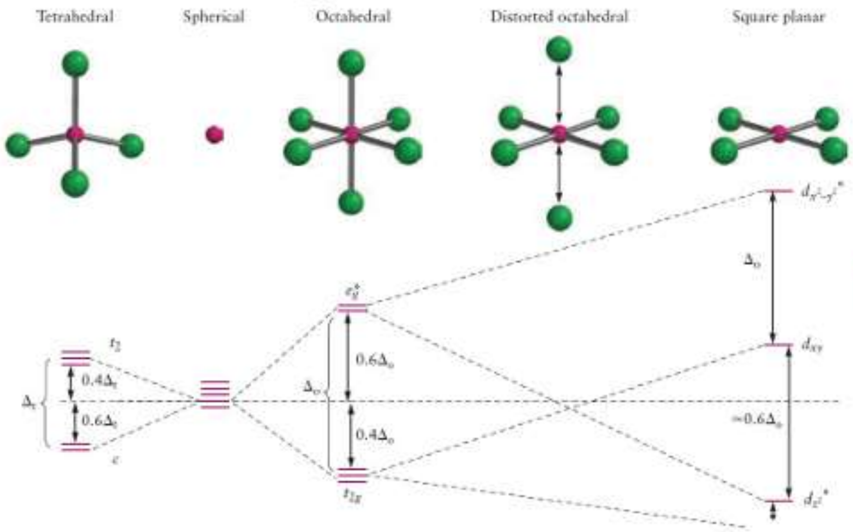


Conjugate system (共轭体系)



Esters羧酸酯

Bonding in transition metal compounds/ coordination complexes



Octahedral: (most common):

- the formation of 6 bonds, rather than 4, confers greater stability.

Tetrahedral: less stable

- The crystal field splitting is smaller
- The lower-energy set of levels is only doubly degenerate

Square-planar: d^8 ions & strong field ligands.

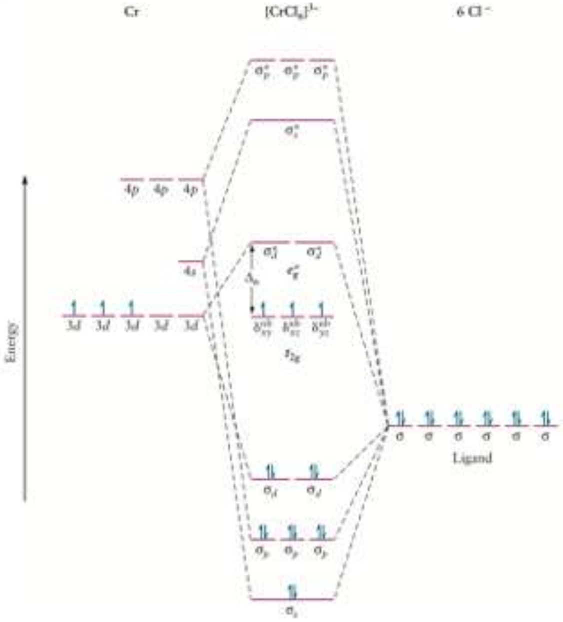
- The low-spin configuration leaves the high-energy d_{x2-y2} orbital vacant is the most stable.

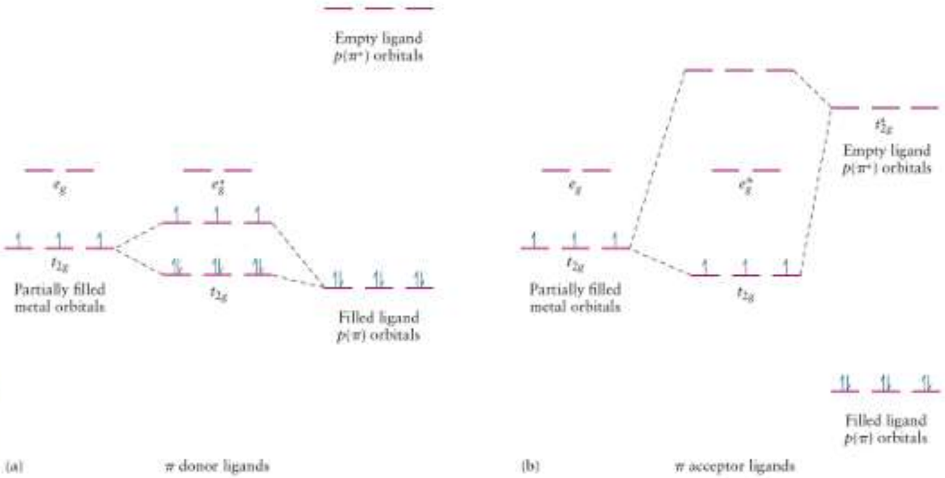
TABLE 8.7

Examples of Hybrid Orbitals and Bonding in Complexes

Coordination Number	Hybrid Orbital	Configuration	Examples
2	*sp*	Linear	$[Ag(NH_3)_2]^+$
3	sp^2	Trigonal planar	BF_3, NO_3^-, $[Ag(PR_3)_3]^+$
4	sp^3	Tetrahedral	$Ni(CO)_4$, $[MnO_4]^-$, $[Zn(NH_3)_4]^{2+}$
4	dsp^2	Square planar	$[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_4]^{2+}$
5	dsp^3	Trigonal bipyramidal	TaF_5, $[CuCl_5]^{3-}$, $[Ni(PEt_3)_2Br_3]$
6	d^2sp^3	Octahedral	$[Co(NH_3)_6]^{3+}$, $[PtCl_6]^{2-}$

From G.E. Kimball, Directed valence. *J. Chem. Phys.* 1940, 8, 188.





$$I^- < Br^- < Cl^- < F^- < OH_2 < NH_3 < en < CO, CN^-$$

π donors σ donors π acceptors

HSAB Theory

Classification of Lewis Acids and Bases†

	Hard	Borderline	Soft
Acids	H^+, Li^+, Na^+, K^+ Be^{2+}, Mg^{2+}, Ca^{2+} Cr^{2+}, Cr^{3+}, Al^{3+} SO_3, BF_3	Fe^{2+}, Co^{2+} Ni^{2+} Cu^{2+} Zn^{2+} Pb^{2+} SO_2, BBr_3	Cu^+, Ag^+ Au^+, Tl^+, Hg^+ Pd^{2+}, Cd^{2+}, Pt^{2+} HG^{2+} BH_3
Bases	F^-, OH^- H_2O, NH_3 CO_3^{2-}, NO_3^-, O^{2-} SO_4^{2-}, PO_4^{3-}, ClO_4^-	$\underline{N}O_2^-$, SO_3^{2-}, Br^- N_3^-, N_2 C_6H_5N, $SC\underline{N}^-$	H^-, R^-, $\underline{C}N^-$, $\underline{C}O$, I^- CN^-, R_3P, C_6H_6 R_2S

Hard acid (硬酸): High charge density: small, multiply charged ion
Soft acid (软酸): Low charge density: large, low oxidation state

'Hard' applies to species which are small, have high charge states (the charge criterion applies mainly to acids, to a lesser extent to bases), and are weakly polarizable.
'Soft' applies to species which are big, have low charge states and are strongly polarizable

HSAB Theory

Comparing tendencies of hard acids and bases vs. soft acids and bases

Property	Hard acids and bases	Soft acids and bases
atomic/ionic radius	small	large
oxidation state	high	low or zero
polarizability	low	high
electronegativity (bases)	high	low
HOMO energy of bases[9][10]	low	higher
LUMO energy of acids[9][10]	high	lower (but more than soft-base HOMO)
affinity	ionic bonding	covalent bonding

“硬亲硬,软亲软,软硬交界都不管”

Solubility: AgF >> AgCl > AgBr > AgI ≈ $Ag(CN)_2$ > Ag_2S

CsF > NaF >> LiF > CaF_2

Examples of hard and soft acids and bases

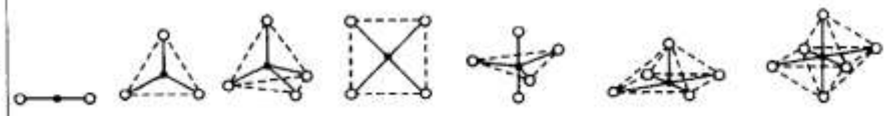
Acids				Bases			
hard		soft		hard		soft	
Hydronium	H_3O^+	Mercury	CH_3Hg^+, Hg^{2+}, Hg_2^{2+}	Hydroxide	OH^-	Hydride	H^-
Alkali metals	Li^+, Na^+, K^+	Platinum	Pt^{2+}	Alkoxide	RO^-	Thiolate	RS^-
Titanium	Ti^{4+}	Palladium	Pd^{2+}	Halogens	F^-, Cl^-	Halogens	I^-
Chromium	Cr^{3+}, Cr^{6+}	Silver	Ag^+	Ammonia	NH_3	Phosphine	PR_3
Boron trifluoride	BF_3	Borane	BH_3	Carboxylate	CH_3COO^-	Thiocyanate	SCN^-
Carbocation	R_3C^+	P-chloranil	$C_6Cl_4O_2$	Carbonate	CO_3^{2-}	Carbon monoxide	CO
Lanthanides	Ln^{3+}	Bulk metals	M^0	Hydrazine	N_2H_4	Benzene	C_6H_6
Thorium, uranium	Th^{4+}, U^{4+}	Gold	Au^+				

Soft acids prefer to form bonds with soft bases, whereas hard acids prefer to form bonds with hard bases, all other factors being equal

Geometry of Coordination Complex

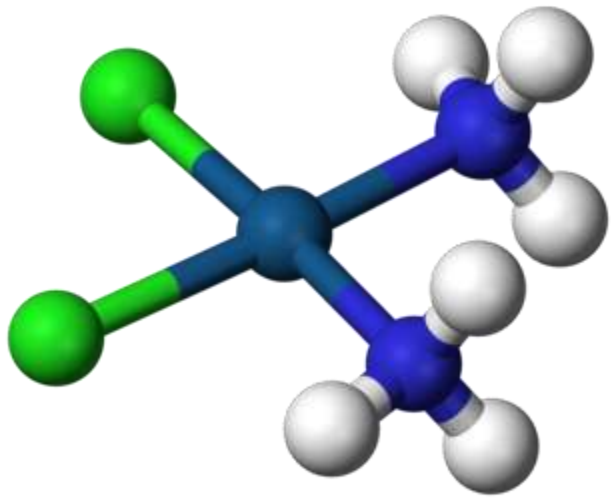
In coordination chemistry, a structure is first described by its coordination number, the number of ligands attached to the metal (more specifically, the number of donor atoms). Usually one can count the ligands attached, but sometimes even the counting can become ambiguous. Coordination numbers are normally between two and nine, but large numbers of ligands are not uncommon for the lanthanides and actinides. The number of bonds depends on the size, charge, and electron configuration of the metal ion and the ligands. Metal ions may have more than one coordination number.

配位数	2	3	4		5		6	
杂化类型	sp	sp^2	sp^3	dsp^2	dsp^3	d^2sp^2	d^2sp^3	sp^3d^2
实 例	$Ag(NH_3)_2^+$	HgI_3^-	$FeCl_4^-$	$PtCl_4^{2-}$	$Fe(CO)_5$	SbF_5^{2-}	$Fe(CN)_6^{4-}$	FeF_6^{3-}
立体构型	直线	平面三角	四面体	平面四边形	三角双锥	四角锥		八面体
模 型								



- Linear for two-coordination
- Trigonal planar for three-coordination
- Tetrahedral or square planar for four-coordination
- Trigonal bipyramidal for five-coordination
- Octahedral for six-coordination
- Pentagonal bipyramidal for seven-coordination
- Square antiprismatic for eight-coordination
- Tricapped trigonal prismatic for nine-coordination

Cisplatin, $PtCl_2(NH_3)_2$, is a coordination complex of platinum(II) with two chloride and two ammonia ligands. It is one of the most successful anticancer drugs.

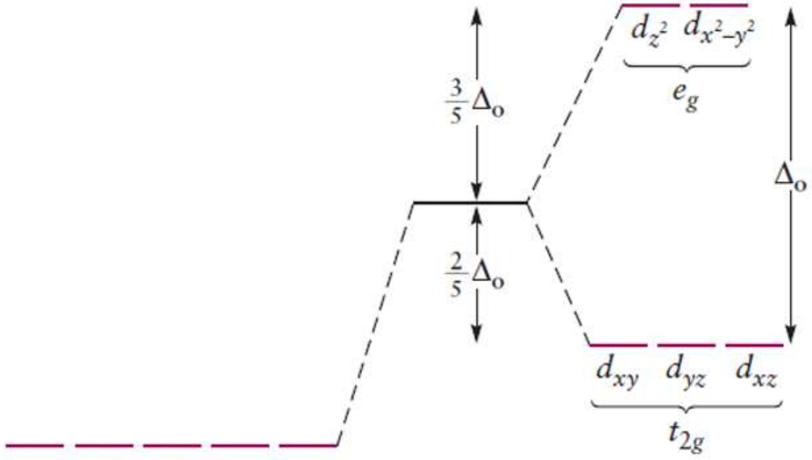


Sp^3d and dsp^2

square-planar geometry: is common for four coordinate complexes of Au^{3+}, Ir^+, Rh^+, and especially common for ions with the d^8 configurations (Ni^{2+} Pd^{2+} and Pt^{2+})

Attention！ Square planar is allowed in coordination chemistry!

Crystal Field Theory

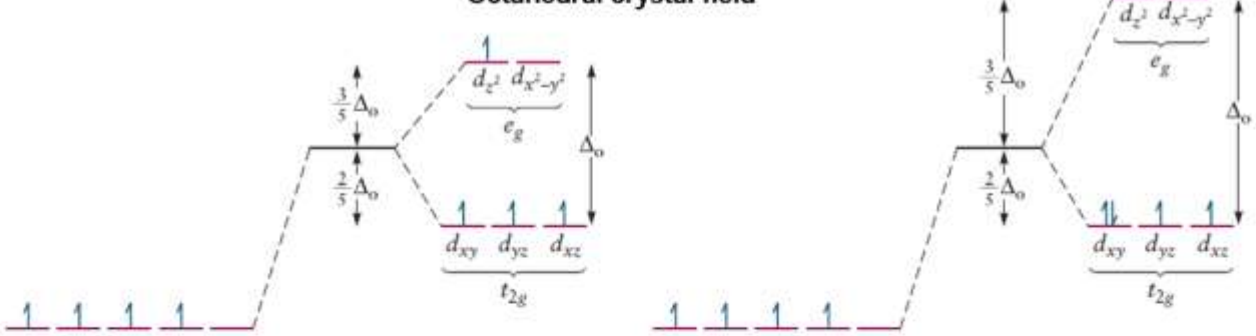


Octahedral crystal field

Δ_o: crystal field splitting energy

In an octahedral field, the energy of the three t_{2g} orbitals is lowered by 2/5 Δ_o, and that of the two e_g orbitals is raised by 3/5 Δ_o relative to the spherical field.

Octahedral crystal field



(a) $[Mn(OH_2)_6]^{3+}$ (high spin) (b) $[Mn(CN)_6]^{3-}$ (low spin)

FIGURE 8.18 Electron configuration for (a) high-spin (Δ_o) and (b) low-spin (δ_o) Mn(III) complexes.

Δ_o is small (weak-field ligands):

- high-spin configurations

Δ_o is large (strong-field ligands)

- low-spin configurations

Electrons are paired in the lower energy t_{2g} orbital, and the e_g orbitals are not occupied until the t_{2g} levels are filled.

The most common type of complex is octahedral, in which six ligands form the vertices of an octahedron around the metal ion. In octahedral symmetry the *d*-orbitals split into two sets with an energy difference, $\boldsymbol{\Delta_{oct}}$ (the crystal-field splitting parameter, also commonly denoted by **10*Dq*** for ten times the "differential of quanta") where the d_{xy}, d_{xz} and d_{yz} orbitals will be lower in energy than the d_{z^2} and $d_{x^2-y^2}$, which will have higher energy, because the former group is farther from the ligands than the latter and therefore experiences less repulsion. The three lower-energy orbitals are collectively referred to as $\mathbf{t_{2g}}$, and the two higher-energy orbitals as $\mathbf{e_g}$.

Crystal Field Theory

表 4－3　八面体配合物的中心原子 d 电子的组态

	弱场		强场	
	d_ε	d_γ	d_ε	d_γ
d^1	↑		↑	
d^2	↑ ↑		↑ ↑	
d^3	↑ ↑ ↑		↑ ↑ ↑	
d^4	↑ ↑ ↑	↑	↑↓ ↑ ↑	
d^5	↑ ↑ ↑	↑ ↑	↑↓ ↑↓ ↑	
d^6	↑↓ ↑ ↑	↑	↑↓ ↑↓ ↑↓	
d^7	↑↓ ↑↓ ↑	↑ ↑	↑↓ ↑↓ ↑↓	↑
d^8	↑↓ ↑↓ ↑↓	↑ ↑	↑↓ ↑↓ ↑↓	↑ ↑
d^9	↑↓ ↑↓ ↑↓	↑↓ ↑	↑↓ ↑↓ ↑↓	↑↓ ↑
d^{10}	↑↓ ↑↓ ↑↓	↑↓ ↑↓	↑↓ ↑↓ ↑↓	↑↓ ↑↓

先确定裂分状态
填入中心原子的d电子（依据强弱场规则）
注意d 4-7 情况
计算能量

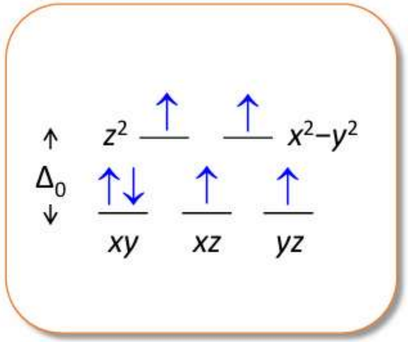
强场配体会倾向于克服电子成对能，将电子“挤”成对

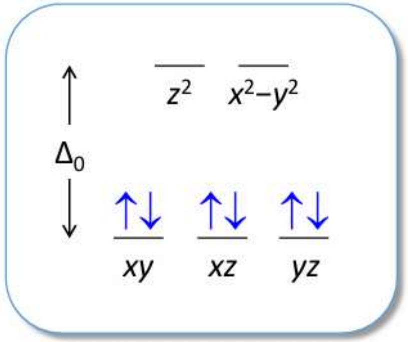
$I^- < Br^- < S_2^- < SCN^- < Cl^- < NO_3^- < N_3^- < F^- < OH^- <$
$OH_2 < NCS^- < py < NH_3 < en < NO_2^- < PPh_3 < CN^- \approx CO \approx NO^+$

Weak-field ligands (high-spin)　Intermediate-field ligands　Strong-field ligands (low-spin)

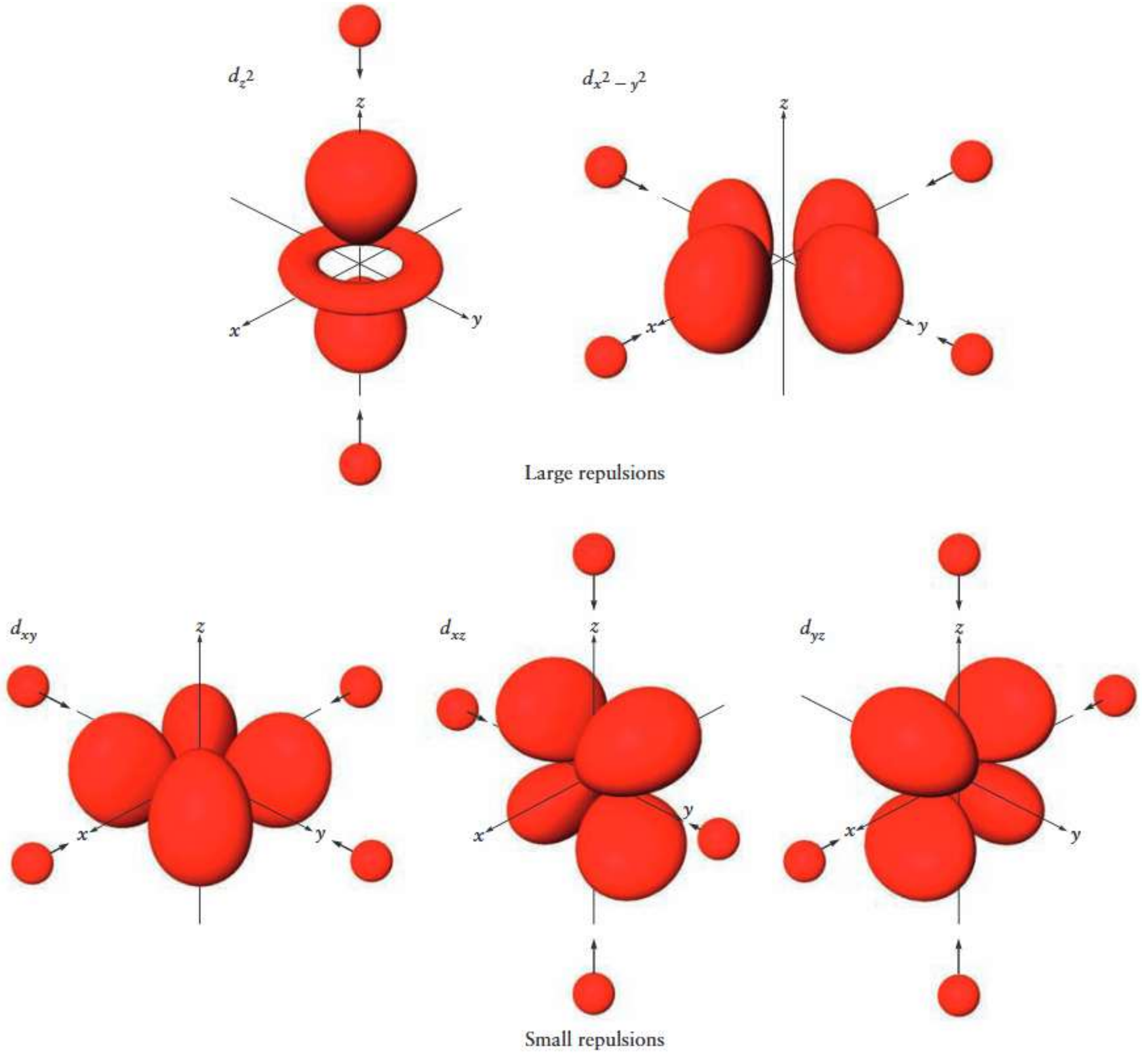
Configuration		d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
Examples		Ti^{3+}	Ti^{2+}, V^{3+}	V^{2+}, Cr^{3+}	Cr^{2+}, Mn^{3+}	Mn^{2+}, Fe^{3+}	Fe^{2+}, Co^{3+}	Co^{2+}, Ni^{3+}	Ni^{2+}, Pt^{2+}	Cu^{2+}	Zn^{2+}
HIGH SPIN	e_g				↑	↑ ↑	↑ ↑	↑ ↑	↑ ↑	↑↓ ↑	↑↓ ↑↓
	t_{2g}	↑	↑ ↑	↑ ↑ ↑	↑ ↑ ↑	↑ ↑ ↑	↑↓ ↑ ↑	↑↓ ↑↓ ↑	↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓
	CFSE	$-\frac{2}{5}\Delta_O$	$-\frac{4}{5}\Delta_O$	$-\frac{6}{5}\Delta_O$	$-\frac{3}{5}\Delta_O$	0	$-\frac{2}{5}\Delta_O$	$-\frac{4}{5}\Delta_O$	$-\frac{6}{5}\Delta_O$	$-\frac{3}{5}\Delta_O$	0
LOW SPIN	e_g							↑			
	t_{2g}				↑↓ ↑ ↑	↑↓ ↑↓ ↑	↑↓ ↑↓ ↑↓	↑↓ ↑↓ ↑↓			
	CFSE	Same as high spin			$-\frac{8}{5}\Delta_O$	$-\frac{10}{5}\Delta_O$	$-\frac{12}{5}\Delta_O$	$-\frac{9}{5}\Delta_O$	Same as high spin		

CFSE, Crystal field stabilization energies.





Crystal Field Theory



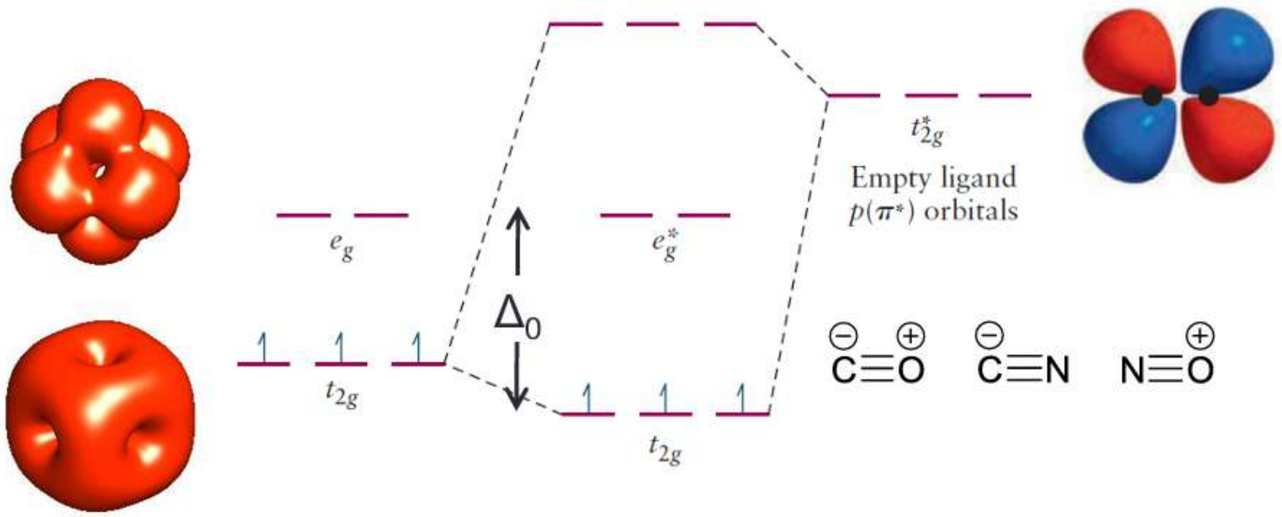
Crystal field theory (CFT) describes the breaking of degeneracies of electron orbital states, usually d or f orbitals, due to a static electric field produced by a surrounding charge distribution (anion neighbors).

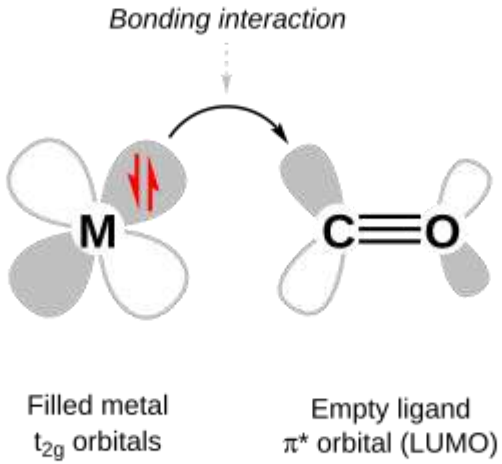
- Tetrahedral
- Octahedral

CUBE model will solve that problem

MO and Crystal Field Theory

Strongest Ligands: π Acceptors





π bonding in octahedral complexes occurs in two ways: via any ligand *p*-orbitals that are not being used in σ bonding, and via any π or π* molecular orbitals present on the ligand.

One important π bonding in coordination complexes is metal-to-ligand π bonding, also called π backbonding. It occurs when the LUMOs (lowest unoccupied molecular orbitals) of the ligand are anti-bonding π* orbitals. These orbitals are close in energy to the d_{xy}, d_{xz} and d_{yz} orbitals, with which they combine to form bonding orbitals (i.e. orbitals of lower energy than the aforementioned set of *d*-orbitals). The corresponding anti-bonding orbitals are higher in energy than the anti-bonding orbitals from σ bonding so, after the new π bonding orbitals are filled with electrons from the metal *d*-orbitals, Δ_O has increased and the bond between the ligand and the metal strengthens. The ligands end up with electrons in their π* molecular orbital, so the corresponding π bond within the ligand weakens.

Molecular spectroscopy and photochemistry

Born–Oppenheimer approximation: slow nuclei, fast electrons. $E_{total}=E_{trans}+E_{rot}+E_{vib}+E_{el}$

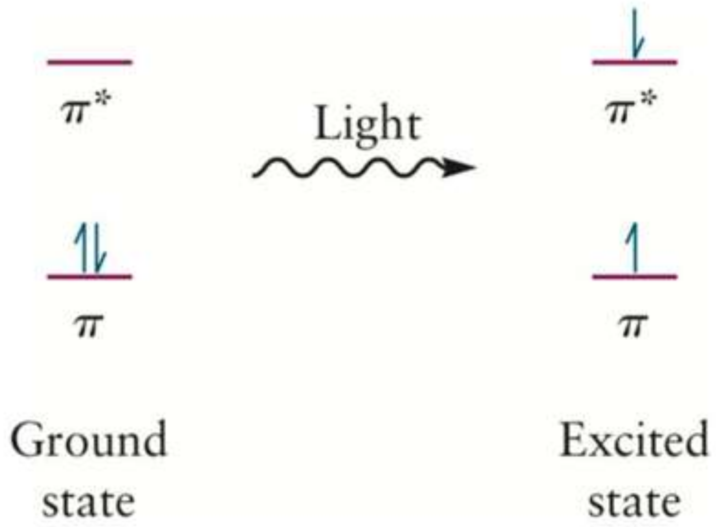
Spectroscopy (光谱): the interactions between electromagnetic radiation and matter.

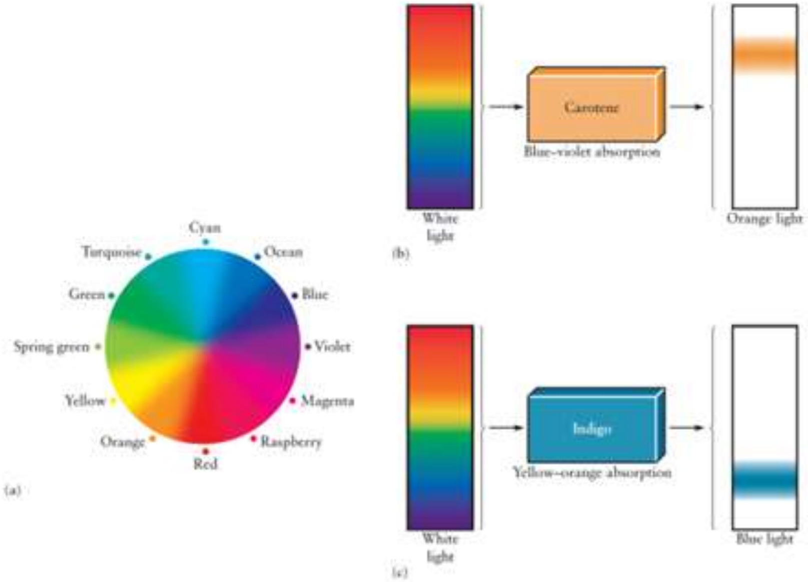
Absorbed 吸收 **Emitted 发射** **Scattered 散射**

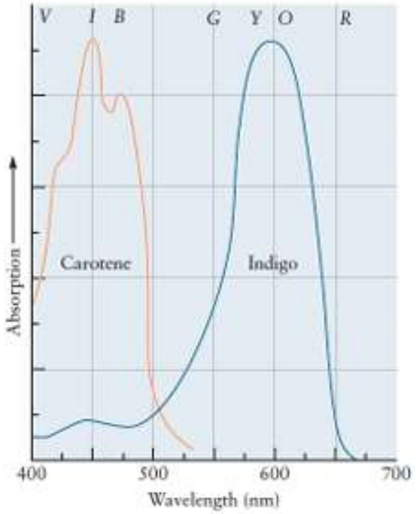
$$\Delta E = h\nu$$

$$\Delta E = h\nu_i - h\nu_s$$

The intensity of the transmitted beam: Beer–Lambert law







“Colors” of complexes

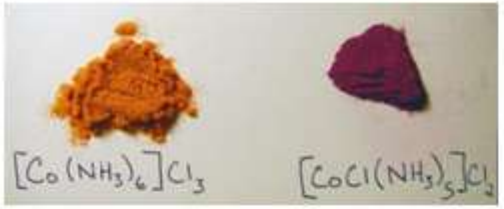
Same chemical formulas
Why different properties?

Compound 1: $CoCl_3 \cdot 6NH_3$ Orange-yellow

Compound 2: $CoCl_3 \cdot 5NH_3$ Purple

Compound 3: $CoCl_3 \cdot 4NH_3$ Green

Compound 4: $CoCl_3 \cdot 3NH_3$ Green



Formulas may cheat
you sometimes

Compound 1: $CoCl_3 \cdot 6NH_3$	Orange-yellow	$+ AgNO_3 \longrightarrow AgCl$	100%	$[Co(NH_3)_6]Cl_3$
Compound 2: $CoCl_3 \cdot 5NH_3$	Purple		66.7%	$[Co(NH_3)_5Cl]Cl_2$
Compound 3: $CoCl_3 \cdot 4NH_3$	Green		33.3%	$[Co(NH_3)_4Cl_2]Cl$
Compound 4: $CoCl_3 \cdot 3NH_3$	Green		0%	$[Co(NH_3)_3Cl_3]$

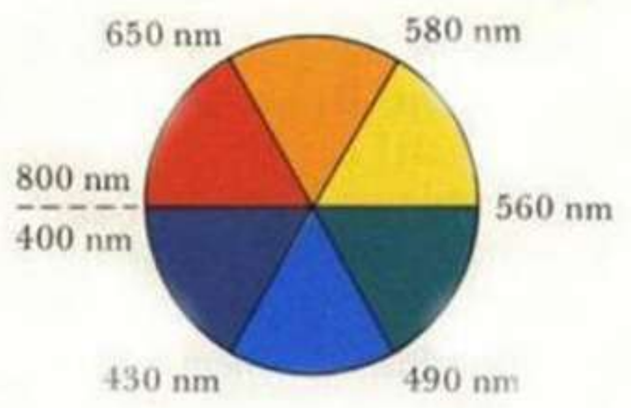
Werner: two different kinds of species associated with the cobalt ions

The primary valences (主价) were anions like Cl^- in simple salts such as $CoCl_3$.
The primary valences were nondirectional. (outer coordination sphere 外界)
The secondary valences (副价) could be either simple anions or neutral molecules such as NH_3.
The secondary valences were oriented along well-defined directions. (inner coordination sphere 内界)

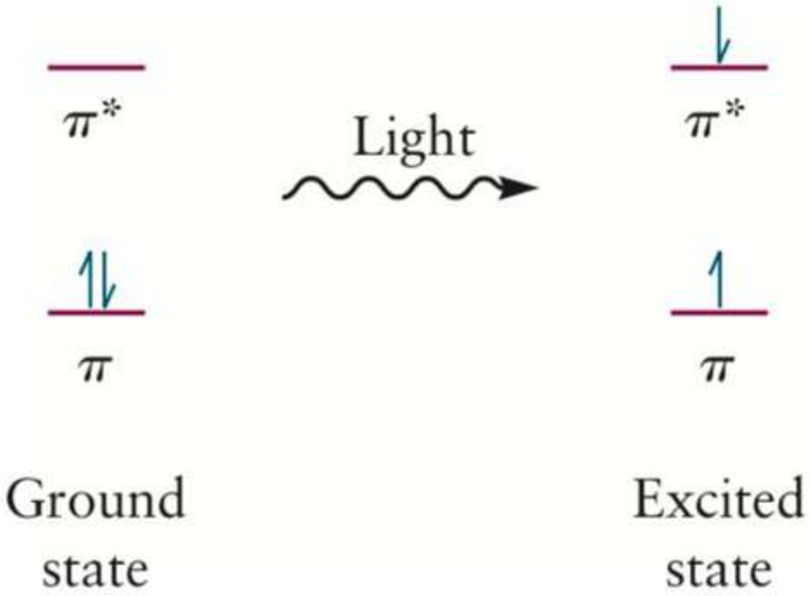
"Colors" of complexes

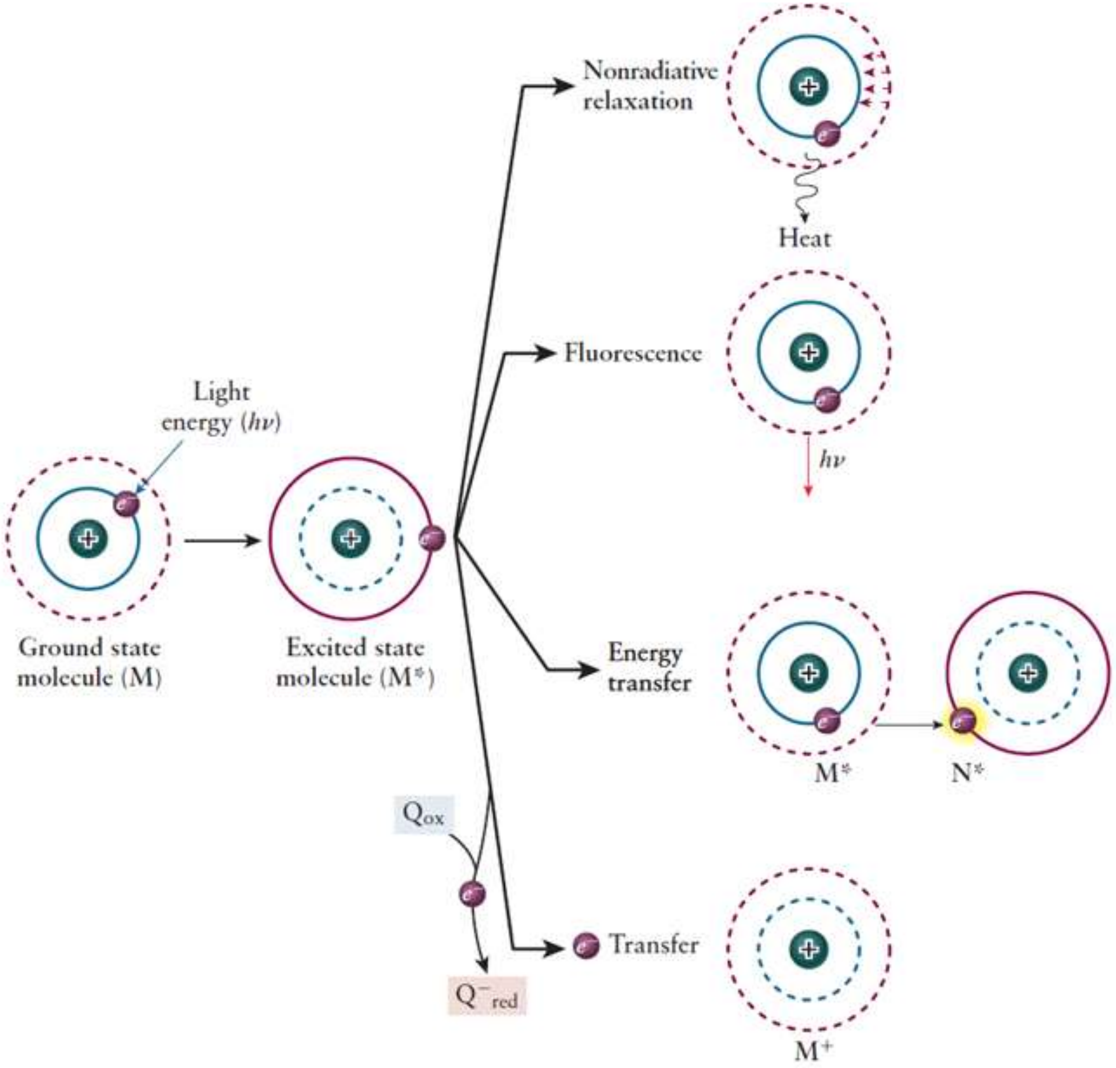
Colours of Various Example Coordination Complexes

	Fe^{2+}	Fe^{3+}	Co^{2+}	Cu^{2+}	Al^{3+}	Cr^{3+}
Hydrated Ion	$[Fe(H_2O)_6]^{2+}$ Pale green Solution	$[Fe(H_2O)_6]^{3+}$ Yellow/brown Solution	$[Co(H_2O)_6]^{2+}$ Pink Solution	$[Cu(H_2O)_6]^{2+}$ Blue Solution	$[Al(H_2O)_6]^{3+}$ Colourless Solution	$[Cr(H_2O)_6]^{3+}$ Green Solution
$(OH)^-$, dilute	$[Fe(H_2O)_4(OH)_2]$ Dark green Precipitate	$[Fe(H_2O)_3(OH)_3]$ Brown Precipitate	$[Co(H_2O)_4(OH)_2]$ Blue/green Precipitate	$[Cu(H_2O)_4(OH)_2]$ Blue Precipitate	$[Al(H_2O)_3(OH)_3]$ White Precipitate	$[Cr(H_2O)_3(OH)_3]$ Green Precipitate
$(OH)^-$, concentrated	$[Fe(H_2O)_4(OH)_2]$ Dark green Precipitate	$[Fe(H_2O)_3(OH)_3]$ Brown Precipitate	$[Co(H_2O)_4(OH)_2]$ Blue/green Precipitate	$[Cu(H_2O)_4(OH)_2]$ Blue Precipitate	$[Al(OH)_4]^-$ Colourless Solution	$[Cr(OH)_6]^{3-}$ Green Solution
NH_3, dilute	$[Fe(NH_3)_6]^{2+}$ Dark green Precipitate	$[Fe(NH_3)_6]^{3+}$ Brown Precipitate	$[Co(NH_3)_6]^{2+}$ Straw coloured Solution	$[Cu(NH_3)_4(H_2O)_2]^{2+}$ Deep blue Solution	$[Al(NH_3)_3]^{3+}$ White Precipitate	$[Cr(NH_3)_6]^{3+}$ Purple Solution
NH_3, concentrated	$[Fe(NH_3)_6]^{2+}$ Dark green Precipitate	$[Fe(NH_3)_6]^{3+}$ Brown Precipitate	$[Co(NH_3)_6]^{2+}$ Straw coloured Solution	$[Cu(NH_3)_4(H_2O)_2]^{2+}$ Deep blue Solution	$[Al(NH_3)_3]^{3+}$ White Precipitate	$[Cr(NH_3)_6]^{3+}$ Purple Solution
$(CO_3)^{2-}$	$FeCO_3$ Dark green Precipitate	$Fe_2(CO_3)_3$ Brown Precipitate+bubbles	$CoCO_3$ Pink Precipitate	$CuCO_3$ Blue/green Precipitate		



“Colors” of complexes

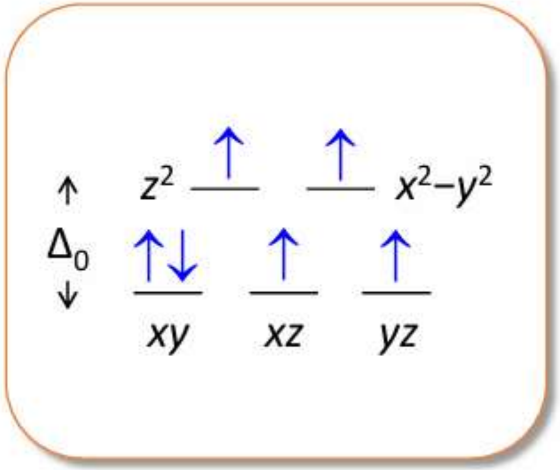


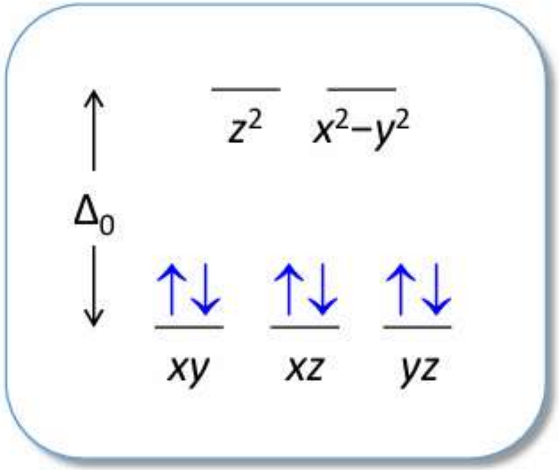


Spectrochemical Series

Optical Properties and the Spectrochemical Series

The wavelength of the strongest absorption is called λ_{max} and it is related to Δ_o





$$\Delta_o = h\nu = hc/\lambda_{max}$$

$[Co(NH_3)_6]^{3+}$	Orange
$[Co(NH_3)_4Cl_2]^{+}$	A green form and a violet form
$[Co(NH_3)_5(H_2O)]^{3+}$	Purple

High spin | Low spin

$I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < CN^- \approx CO$

The smaller halide ions can approach the metal ion more closely: greater electrostatic repulsion.

End

- Chemistry is a kind of fundamental science.
- Know the structure better, then you can know your world better.
- Try to read English materials and textbooks even though it’s not easy for beginners.
- Collaboration makes perfect.

……….

- Good to have you all.
- Thank you.

Best wishes~